C-CLARKE
INTERNATIONAL INSTITUTE OF DIGITAL SCIENCES

C-CLARKE INTERNATIONAL INSTITUTE OF DIGITAL SCIENCES

Innovating Education for Tomorrow's Leaders



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WELCOME MESSAGE

Dear Students,

Welcome to C-Clarke International Institute of Digital Sciences!

We are thrilled to have you join us on this exciting journey of learning, innovation, and professional growth with us. At C-Clarke, we are dedicated to providing you with an exceptional education that combines advanced technologies, practical skills, and industry insights to prepare you for a successful career in software landscape.

Our institute is more than just a place to study—it's a community where connections are made, ideas are shared, and futures are shaped. Guided by a team of esteemed lecturers and supported by strong industry partnerships, we are committed to helping you achieve your goals and become leaders who make meaningful contributions both locally and globally.

We encourage you to take full advantage of the resources, opportunities, and support available at C-Clarke. Together, let's build a brighter future for you and our society.

Wishing you great success

Director, Academic Affairs

ABOUT US

C-Clarke International Institute of Digital Sciences is a premier educational institution dedicated to shaping the future of software technology professionals. With a commitment to delivering innovative, industry-relevant education, we enable students to excel in the fast-evolving world of software and digital sciences.

Our programs are precisely designed to provide a strong foundation in advanced technologies, practical skills, and ethical practices, ensuring our students are equipped for success both locally and globally. Backed by a distinguished panel of lecturers and consultants with extensive industry experience, we offer unparalleled opportunities for learning and networking.

At C-Clarke, we prioritize not just education but also the holistic development of our students. By fostering innovation, growth, and lifelong learning, we aim to create a community of future leaders who drive progress, achieve economic independence, and contribute meaningfully to the world.

C-Clarke is not just a place to study; it's a place to connect, grow, and transform your aspirations into reality.



VISION AND MISSION

Vision
To Be acknowledged as the Leader in Education for Digital Sciences, Whilst Inspiring Young Leaders to Achieve Economic Strength and Independence.

Mission
To Provide Innovative Education That Builds Essential Skills for Professional Success Locally and Globally.

CORE VALUES

Innovation
Continuously adhering to the latest global technologies, ensuring our programs stay ahead of industry trends to prepare students for the future.

Integrity
Committed to transparency, honesty, and ethical practices, fostering trust and respect in all our educational endeavors.

Excellence
Striving for the highest standards in education and guiding students to reach their full potential through world-class learning experiences.

Empathy
Building a supportive and compassionate community that understands the personal and professional aspirations of each student, guiding them with care and dedication toward their success

PANEL OF LECTURERS AND ACADEMIC CONSULTANTS



DR. SOORIYA BANDARA

PhD in Electrical & Electronics Engineering,
NTU Singapore
BSc (Hons) in Electrical and Electronic Engineering
University of Peradeniya

Staff Engineer,
Wireless & Contact Charging Design at **Sivantos Group**
Visiting Lecturer at University of Peradeniya



KUSHANTHA GUNAWARDANA

PhD (Reading)
RMIT University
Cyber/ Computer Forensics and Counter Terrorism
MSc-IT
University of Colombo
MCT, C|EI, C|PENT, LPT, E|CSA, C|EH, C|HFI, E|NSA, C|ND, E|CIH, CCNSP, IRNSS, CISE, MIEEE, MBCS, MCP, MCSA +M, MCSE, HNDIT

Microsoft Cyber Security Architect
Expert International Cybersecurity Trainer & Practitioner
Founder/ CTO - Zecureright Pty Ltd (Australia)



GAYAN GUNAWARDANA

BSc in IT
University of Moratuwa
MBA, Marketing
University Of Belfordshire
Expert in Identity & Access Management
CISSP certified | AWS certified

Founder - oneIAM
IAM & Information Security Experts



LIMANTHA MALIKA

BSc (Hons) in
Computer Science & Technology
Uva Wellassa University
MSc in IT
University of Colombo
IsTQB Certified Tester

Lead Engineer
Software Quality Assurance
HCL Technologies

PANEL OF LECTURERS AND ACADEMIC CONSULTANTS



PRASHAN SILVA

Bsc (Hons) in Computer Science
University Of Kelaniya

Tech Lead at CodeGen



SUSHAIN DILISHAN PATHIRAGE

BSc (Hons) in Software Engineering
Cardiff Metropolitan University

Expert in Full Stack Development

Technical Lead
Sysco LABS



DILSHAN ARIYARATHNA

BSc (SP) in Computer Science
University of Kelaniya

Associate Technical Lead Engineer at iCan Lanka



CHAMINDIKA KODITHUWAKKU

Master of Information Technology
University of Colombo
BSc. (Hons) in IT & Management
University of Moratuwa

Senior software engineer
Ambrum Solutions (pvt) Ltd

ADVANCED AI & SOFTWARE ENGINEERING

Advanced AI & Professional Software Engineering is a concise, yet intensive, 12-month program that focuses on foundational technologies and tools essential for any aspiring software developer.

Future with AI

After mastering advanced Python development as your foundation, then we dive deep into the world of Generative AI. You can learn to integrate powerful AI APIs like GPT, Claude, LLaMA, and Perplexity to build your own intelligent tools. You'll also gain knowledge and experience in creating multi-agent AI systems and mastering prompt engineering — the skill of communicating effectively with large language models. We help you not only just to understand AI, but also to create with it.

Real-World Benefits:

This program is designed to quickly bring you up to speed with the core competencies required in the software industry. You'll gain hands-on experience in building web applications, managing code through version control, and developing a solid understanding of key programming languages. Graduates are well-prepared to take on entry-level roles in software development or enhance their current skill set.

FULL COURSE SYLLABUS

Module 1: Introduction to Java Programming

1.1 Compiler
1.2 Interpreter
1.3 Java Fundamentals
1.4 Data Structures and Algorithms

Module 2: Introduction to Web Development

2.1 HTML5 / CSS3
2.2 JavaScript ES6
2.3 Ajax

Libraries:
- 2.4 Bootstrap
- 2.5 Tailwind
- 2.6 jQuery

Module 3: OOP & Design Patterns

3.1 Object-Oriented Programming in Java
- a. Introduction to OOP
- b. Encapsulation
- c. Inheritance
- d. Polymorphism
- e. Abstraction

3.2 Design Patterns

Module 4: Database Management Systems

4.1 Database Models & Architecture
4.2 Relational Database Concepts
4.3 SQL
4.4 Advanced SQL & Performance Optimization
4.5 NoSQL Databases
4.6 Database Design & Modeling
4.7 Database Security & Administration

Module 5: Advanced Java Development

5.1 Java Threads
5.2 Synchronization in Java
5.3 Exception Handling
5.4 File Handling
5.5 Lambda Expression

Module 6: Comprehensive React Training

6.1 Introduction to React
6.2 JavaScript XML
6.3 Event Handling
6.4 Routing and Navigation
6.5 React Lifecycle
6.6 State Management
6.7 React Hooks
6.8 Functional Components
6.9 Higher Order Components

Module 7: Spring & Spring Cloud

7.1 Introduction to Spring Framework
7.2 Building RESTful Web Services
7.3 Spring Data JPA
7.4 Spring Boot Security
7.5 Microservices with Spring Boot & Spring Cloud

Module 8: Professional Python Development

8.1 Python Basics
8.2 Advanced Data Structures
8.3 Control Flow & Loops
8.4 OOP
8.5 Data Structures
8.6 Modules & Packages
8.7 Functions & Exception Handling
8.8 File Handling & Exceptions
8.9 Regular Expressions & Debugging

Module 9: Comprehensive Angular Training

9.1 Introduction to Angular
9.2 Components and Templates
9.3 Databinding and Directives
9.4 Services and Dependency Injection
9.5 Routing and Navigation
9.6 HTTP Client and Observables
9.7 Pipes
9.8 Testing and Debugging

Module 10: Advanced AI Development

10.1 Mathematics for AI
10.2 Introduction to AI API
10.3 Data Handling and Processing
10.4 Machine Learning Fundamentals
10.5 Deep Learning with AI API
10.6 Real-World AI Applications
10.7 Natural Language Processing
10.8 Ethics and AI
10.9 AI DevOps and MLOps

Module 11: Prompt Engineering

11.1 LLM Concepts
11.2 Introduction to Prompting
11.3 Prompting Techniques
11.4 Improving Reliability

Module 12: QA, Testing & Automation

12.1 Fundamentals of Testing
12.2 Types of Testing & Test Cases
12.3 Bug Reporting & Test Case Design
12.4 Test Techniques
12.5 Automation

Module 13: Flutter App Development

13.1 Introduction to Flutter
13.2 Dart Programming
13.3 Flutter Basics
13.4 State Management
13.5 Networking and Data Storage

Module 14: Advanced Project Management

14.1 Project Management Frameworks
14.2 Methodologies
14.3 Initiation & Stakeholder Management
14.4 Planning & WBS
14.5 Agile Frameworks

Module 15: Enterprise App Development

15.1 Software Engineering Life Cycle
15.2 Advanced DBMS
15.3 Enterprise App Development Project

Module 16: LLM Engineering

16.1 Overview of Open Source & Proprietary LLMs (GPT, Claude, LLaMA, Mistral, etc.)
16.2 LLM Inference with APIs (OpenAI, HuggingFace, Ollama, etc.)
16.3 LLM Fine-Tuning vs Prompt Engineering
16.4 RAG Architecture (Retrieval-Augmented Generation)
16.5 LangChain & LlamaIndex Integration
16.6 Custom Tool Integration & Function Calling
16.7 Guardrails, Evaluation & Monitoring LLM Apps
16.8 Deploying LLM Applications with Docker & Streamlit
16.9 Building Autonomous AI Agents & Workflows
16.10 LLM Application Project

Module 17: Version Control

17.1 Introduction to Version Control
17.2 Git Basics (init, clone, commit, push, pull)
17.3 Branching and Merging
17.4 Resolving Conflicts
17.5 Working with GitHub Repositories
17.6 Collaboration with GitHub (PRs, Issues, Projects)
17.7 GitHub Actions for CI/CD

Module 18: AWS Cloud & DevOps

18.1 AWS Introduction & IAM
18.2 EC2, S3, and Cloud Storage
18.3 VPC & Networking Basics
18.4 RDS & DynamoDB
18.5 AWS Lambda & Serverless
18.6 DevOps on AWS (CI/CD Pipelines)
18.7 CloudFormation & Infrastructure as Code
18.8 Monitoring & Logging (CloudWatch, X-Ray)
18.9 AWS Deployment Project

LIFETIME LEARNING SUPPORT

We provide:

- Access to updated course materials.
- Regular webinars on industry trends.
- A dedicated alumni network.

ADMISSION PROCESS

» Visit our website and fill out the application form.
» Submit necessary documents.
» Attend an interview or placement test.
» Confirm your admission and begin your journey.

COURSE FEES AND PAYMENT OPTIONS

Bank Account Details for Payments

COMMERCIAL BANK

Account Name: C-Clarke International Institute of Digital Sciences
Bank: Commercial Bank **Bank Code:** 7056
Branch: Moratuwa Branch **Code:** 59
Account Type: Current Account
Account Number: 1000874493

Please Note:

- **All completed payments are non-refundable**
- **We encourage all students to reach out for guidance to ensure a smooth payment process.**

PAN ASIA BANK

Account Name: C-Clarke International Institute of Digital Sciences
Bank: Pan Asia Bank **Bank Code:** 7311
Branch: Moratuwa Branch **Code:** 024
Account Type: Current Account
Account Number: 102411001581

Important Instructions

- When making the payment, please include your NIC number on the deposit slip or in the remarks section for online transfers.
- If you require specific payment instructions, contact your Student Counsellor before proceeding:
- Hotline: 070 1000000

ADVANCED AI & SOFTWARE ENGINEERING PROGRAM

Course Fee: 180,000 LKR
Special Offer: 40,000 LKR Off For One-Time Payments
- One-Time Fee: 140,000 LKR

Duration : 12 Months

> **STEP INTO**
> **Future AI Development with Advanced AI & Software Engineering Program**



REFUND POLICY

At C-Clarke International Institute of Digital Sciences, we maintain a strict **NO-REFUND POLICY** for all completed payments. Once a payment is made, it is considered final and non-refundable.

We strongly encourage all prospective students to review the program details, schedules, and payment terms carefully before proceeding with the payment. For any clarifications, our student counselors are available to assist you prior to confirming your enrollment.

This policy ensures the smooth operation of our programs and the continued delivery of high-quality education to all enrolled students.

CONTACT INFORMATION

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Facebook	:	http://www.facebook.com/cclarkecampus
Instagram	:	http://www.instagram.com/cclarkecampus